EXAMINATIONS COUNCIL OF ZAMBIA

Examination for General Certificate of Education Ordinary Level

Additional Mathematics 4030/1

Paper 1

2021

Additional Materials:
Answer Booklet
Silent electronic calculator (Non programmable)

Time: 2 hours **Marks: 80**

Instructions to Candidates

1 Write the **centre number** and your **examination number** on **every page** of the separate **Answer Booklet** provided.
2 There are **twelve** questions in this paper. Answer **all** questions.
3 Write your answers in the separate Answer Booklet provided.
4 If you use more than one Answer Booklet, fasten the Answer Booklets together.
5 Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Information for Candidates

1 The number of marks is shown in brackets [ ] at the end of each question or part question.
2 **The use of a non programmable electronic calculator is expected, where appropriate.**
3 You are reminded of the need for clear presentation in your answers.
4 Cell phones and other electronic devices are **not allowed** in the examination room.
5 Check the formulae overleaf.

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This question paper consists of 5 printed pages

4030-1-000340

MATHEMATICS FORMULAE

1 ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots\ldots + \binom{n}{r} a^{n-r} b^r + \ldots\ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

2 TRIGONOMETRY

Identities

$\text{Sin}^2 A + \cos^2 A = 1$

$\text{Sec}^2 A = 1 + \tan^2 A$

$\text{Cosec}^2 A = 1 + \cot^2 A$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$a^2 = b^2 + c^2 - 2bc \cos A$

$\Delta = \frac{1}{2} bc \sin A$

1 Given that the coordinates of the midpoint of the points $(1 - p, 1 + t)$ and $(2p, 4t + 3)$ are $(3, 7)$, find the value of p and of t. [4]

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2 Solve the simultaneous equations

$2y = x + 5$,

$x^2 + xy = -1$. [5]

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3 Find the values of k for which $x + 2y = k$ is a tangent to the curve $y^2 + 4x - 20 = 0$. [4]

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4 The functions f and g are defined by $f: x \mapsto \frac{x+5}{2x}$, $x \neq 0$ and $g: x \mapsto 2x + 4$,

where x is a real number.

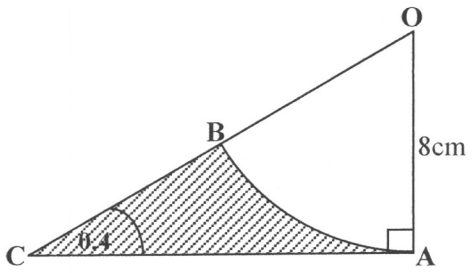
Find

(a) $(fg)^{-1}(x)$, [4]

(b) the value of x for which $(fg)^{-1}(x) = -3$. [2]

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5 In the diagram below, triangle OCA is right angled at A, angle ACO is 0.4 radians and OBA is a sector of a circle centre O and radius 8 cm.



Calculate

(a) the length of AC, [2]

(b) the area of the shaded region. [3]

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6 Prove the identity

$$\frac{1}{(\sec\theta - \tan\theta)^2} \equiv \frac{1+\sin\theta}{1-\sin\theta}.$$ [4]

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7 **(a)** Find the coefficient of x in the expansion of $\left(x^2 - \frac{4}{x}\right)^5$. [4]

(b) Obtain the first four terms in the expansion of $(1 + p)^8$ in ascending powers of p. Hence find the coefficient of x in the expansion of $(1 + x + 2x^2)^8$. [5]

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8 **(a)** Find the gradient function of $y = \dfrac{2x^2}{3x^2 + 6}$. [3]

(b) Find the equation of the normal to the curve $y = \dfrac{3}{2x - 5}$, $x \neq \dfrac{5}{2}$, at a point where $x = 2$. [6]

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9 On the same diagram, sketch the graphs of $y = \left|\cos\left(\dfrac{x}{2}\right) - 1\right|$ and $y = 1$ for the domain $0 \leq x \leq 2\pi$. Hence state the number of solutions for which $\left|\cos\left(\dfrac{x}{2}\right) - 1\right| = 1$. [5]

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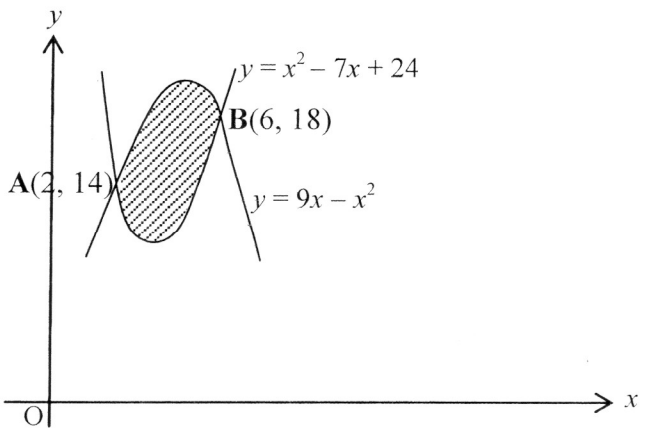
10 **(a)** Find the vector equation of the perpendicular bisector of the line joining the points (–3, –9) and (11, 5). [4]

(b) The position vectors of P, Q and R are $3i - 2j$, $ni + j$ and $2i - 8j$ respectively. Find the value of n for which P, Q and R are collinear. [5]

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11 **(a)** Find $\int\left(x^3 - \dfrac{6}{x^2}\right)dx$. [2]

(b) The diagram below shows part of the curves $y = x^2 - 7x + 24$ and $y = 9x - x^2$ intersecting at A(2, 14) and B(6, 18).



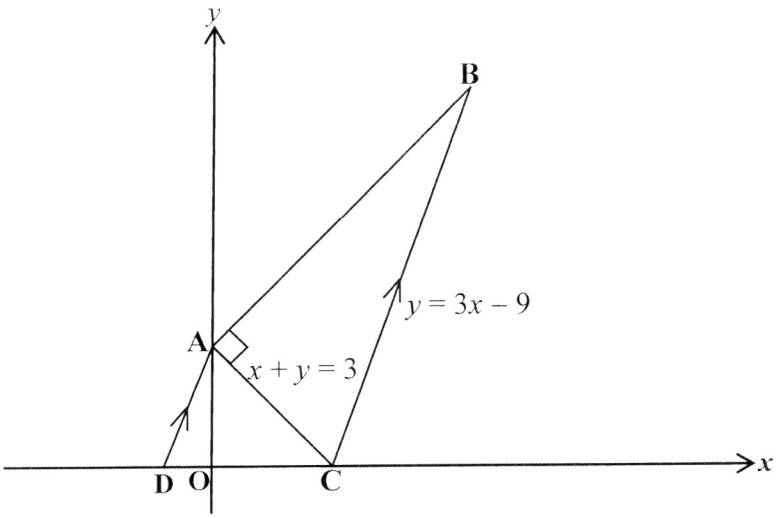
Find

(i) the tangent to the curve $y = x^2 - 7x + 24$ at the point B, [4]

(ii) the area of the shaded region. [4]

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12 In the diagram below, ABCD is a trapezium in which angle BAC = 90° and AD is parallel to CB. The points C and D lie on the x-axis and A lies on the y-axis. The equations of lines CB and AC are $y = 3x - 9$ and $x + y = 3$ respectively.



Find

(a) the equation of line AB, [3]

(b) the coordinates of the points D and B, [4]

(c) the area of the trapezium ABCD. [3]